



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 12

ENGINEERING GRAPHICS AND DESIGN P2

NOVEMBER 2025

MARKING GUIDELINES

MARKS: 100

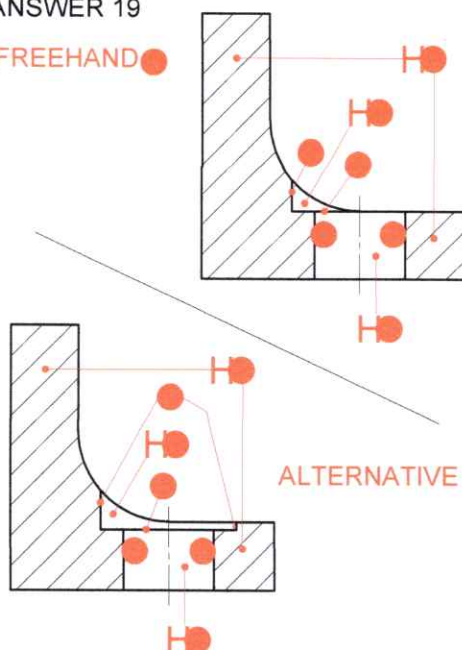
These marking guidelines consist of 7 pages.

ANSWERS

1	JJ-14213	1
2	THOMAS	1
3	CUSTOM CONTAINERS	1
4	AUTOCAD 2024	1
5	www.prodraw.co.za	1
6	THIRD-ANGLE ORTHOGRAPHIC PROJECTION	1
7	150	1
8	RIGHT VIEW	1
9	PARTIAL	1
10	REVOLVED	1
11	ACROSS FLATS	1
12	3	1
13	6°	1
14	F: 28 G: M10 H: 31 J: M6	4
15	MAX - 14.35	1
	MIN - 13.75	1
16	24	1
17	NO MACHINING	1
18	b: <u>PRODUCTION METHOD</u> / <u>TREATMENT</u> / <u>COATING</u> / <u>FINISH</u>	1
	d: <u>DIRECTION OF LAY</u>	1
19	See below	4
20		3
TOTAL		30

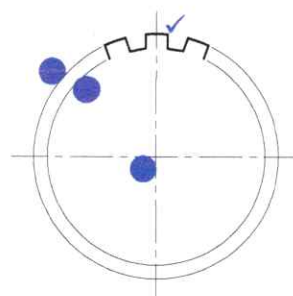
ANSWER 19

FREEHAND

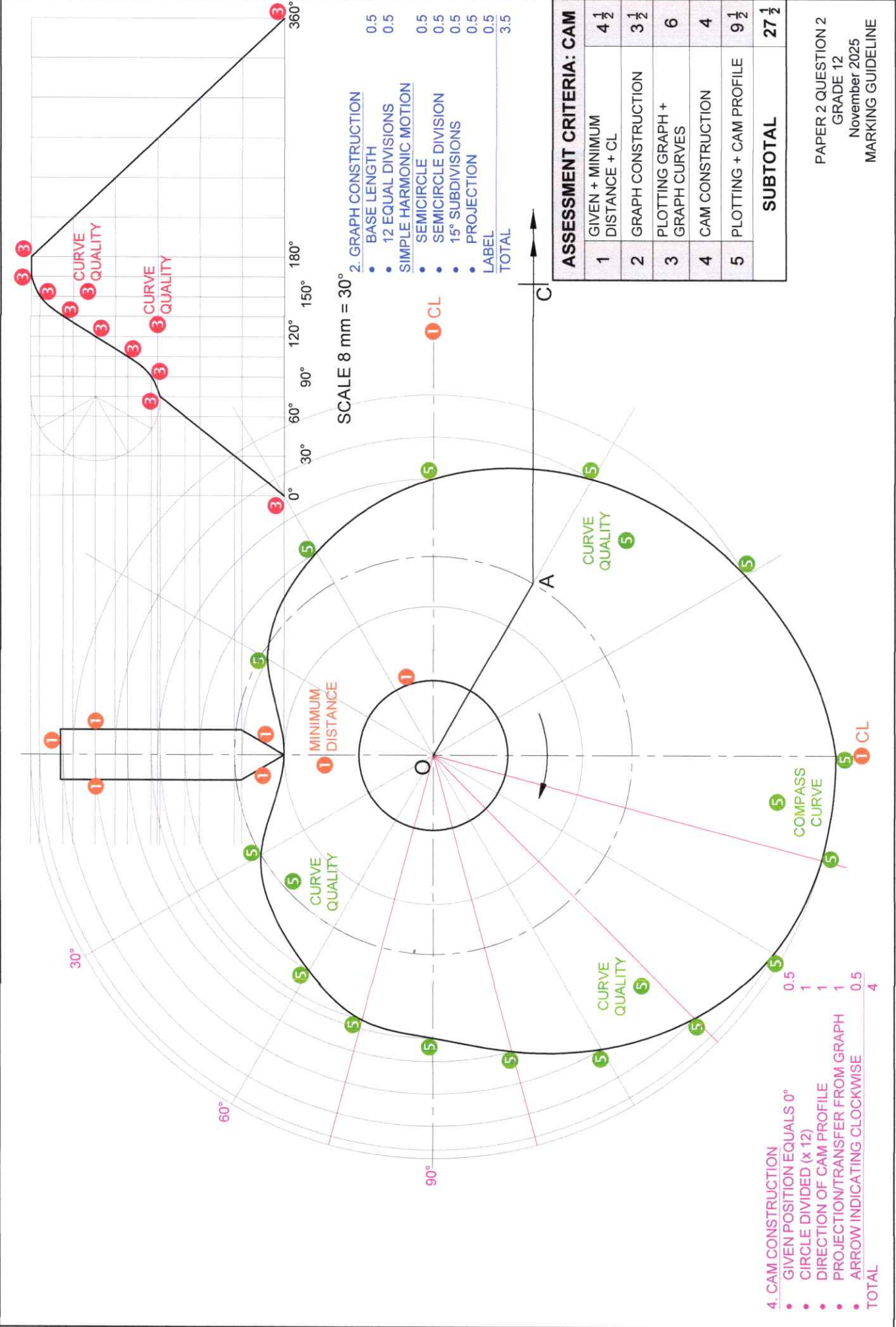


ANSWER 20

FREEHAND



PAPER 2 QUESTION 1
GRADE 12
November 2025
MARKING GUIDELINE





ASSESSMENT CRITERIA: MECHANISM		
1	GIVEN	$3\frac{1}{2}$
2	CONSTRUCTION	3
3	POINTS + CURVE	7
SUBTOTAL		$13\frac{1}{2}$

PAPER 2 QUESTION 2
GRADE 12
November 2025
MARKING GUIDELINE

STARTING 1 / ORIENTATION 1

CIRCLE CONSTRUCTION 4

ASSESSMENT CRITERIA	
1	PLACING + AUX. VIEW
2	FRONT PORTION
3	BACK PORTION
4	CIRCLES + CONSTR. + CL
TOTAL	

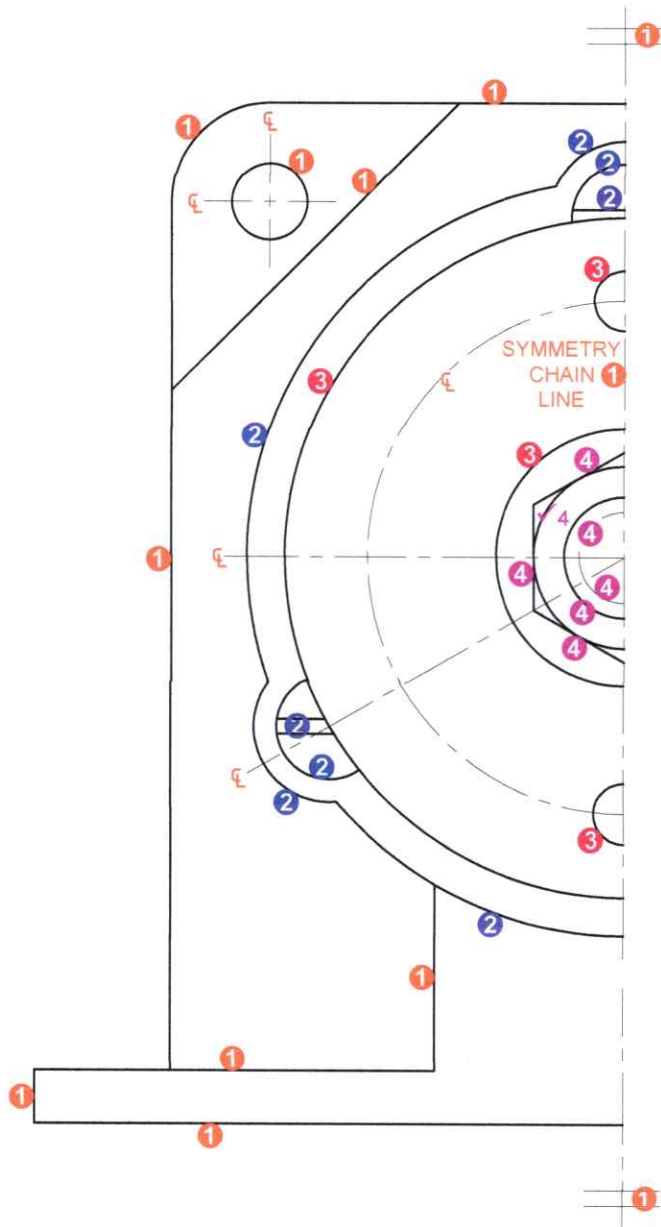
2
14 1/2
14 1/2
8
39

PAPER 2 QUESTION 3

GRADE 12

November 2025

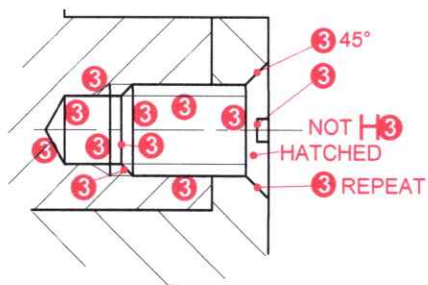
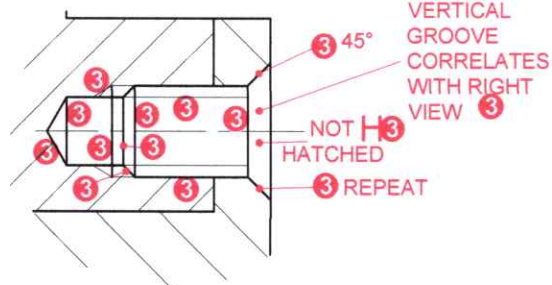
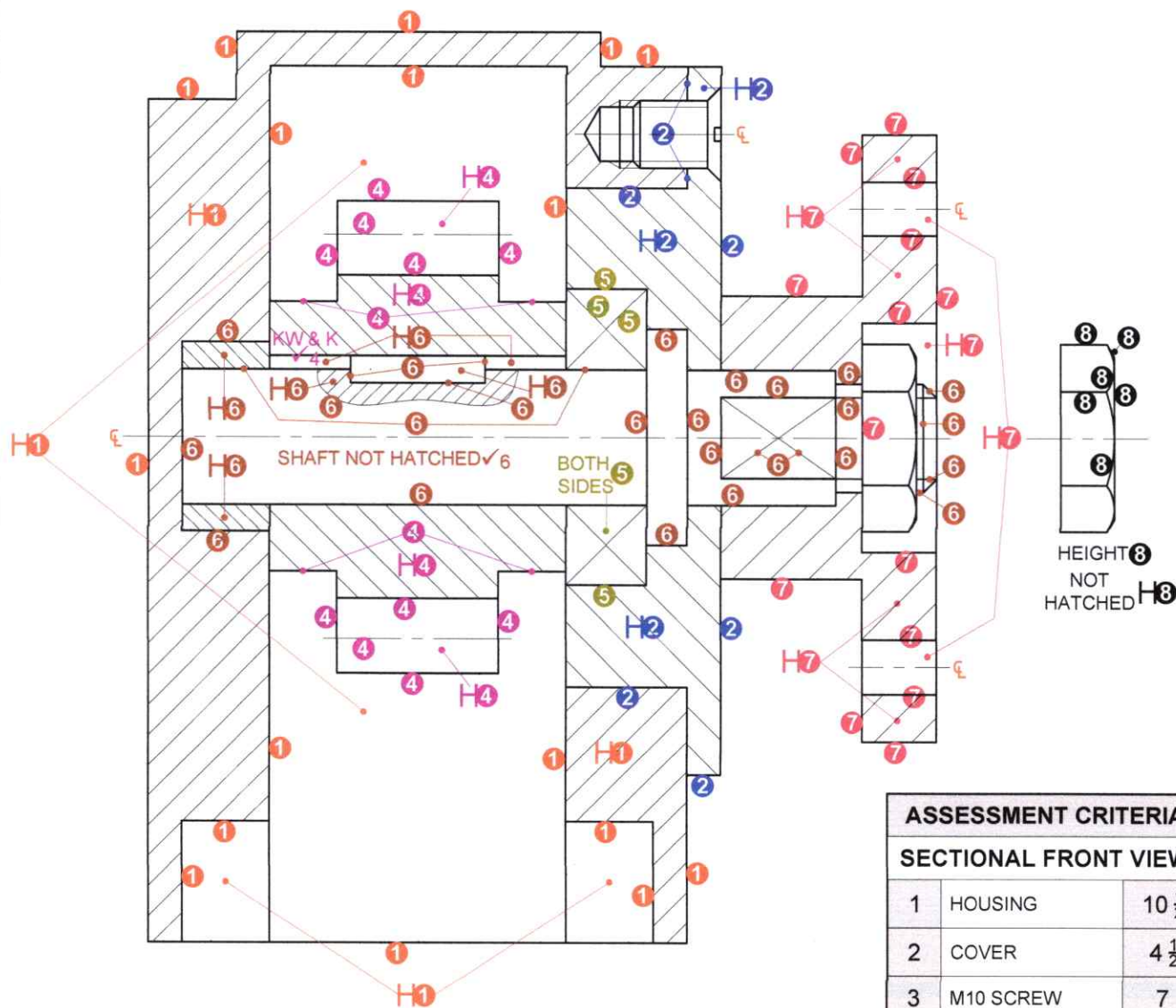
MARKING GUIDELINE



1. CENTRE LINES (CL)
SANS COMPLIANT
CENTRE LINES (1 - 3 = 0,5; 4 - 5 = 1; 6 - 7 = 1,5; 8 - 9 = 2) 1 2 3

ASSESSMENT CRITERIA		
RIGHT VIEW		
1	HOUSING + SYMMETRY	6
2	COVER + SCREWS	4
3	FLANGE COUPLING	2
4	M16 NUT + SHAFT	4
SUBTOTAL		16

PAPER 2 QUESTION 4
GRADE 12
November 2025
MARKING GUIDELINE

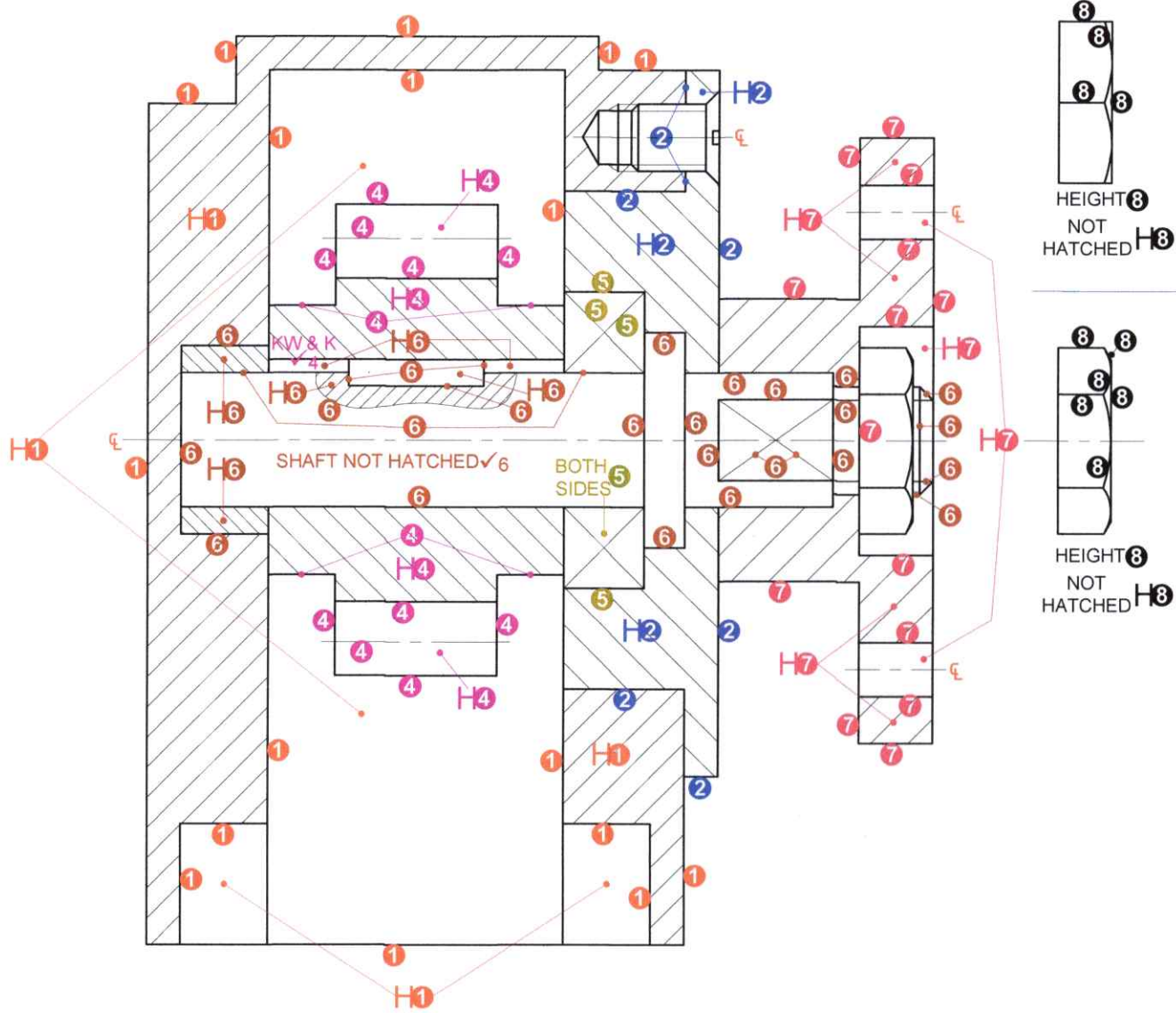
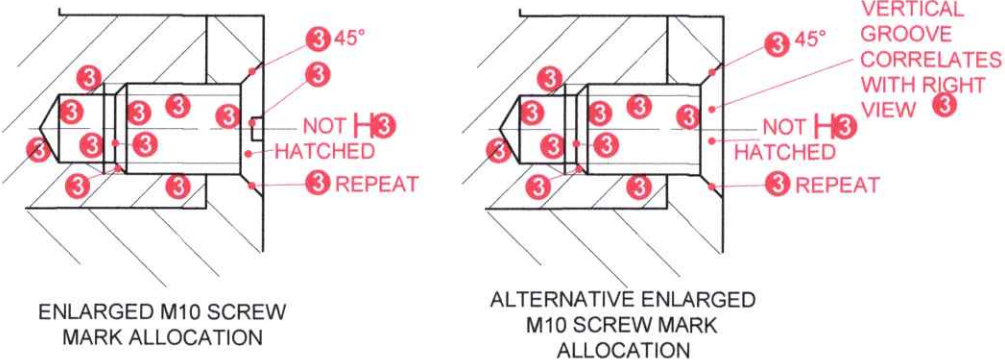
ENLARGED M10 SCREW
MARK ALLOCATIONALTERNATIVE ENLARGED
M10 SCREW MARK
ALLOCATION

2. ASSEMBLY
1 MARK FOR EVERY PART CORRECTLY ASSEMBLED
(10 PARTS - 1) = 9

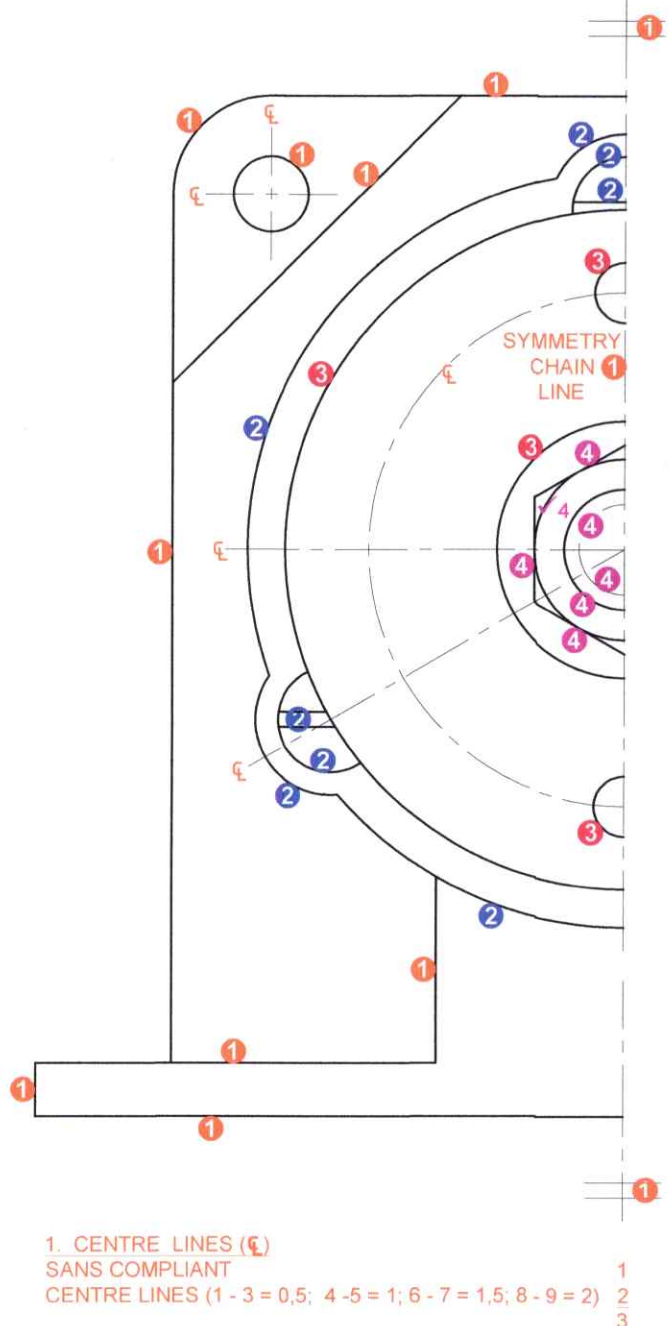
ASSESSMENT CRITERIA**SECTIONAL FRONT VIEW**

1	HOUSING	10 $\frac{1}{2}$
2	COVER	4 $\frac{1}{2}$
3	M10 SCREW	7
4	SPUR GEAR	9
5	BALL BEARING	2 $\frac{1}{2}$
6	SHAFT + KEY + BUSH	15 $\frac{1}{2}$
7	FLANGE COUPLING	9
8	M16 NUT	4
SUBTOTAL		62

PAPER 2 QUESTION 4
GRADE 12
November 2025
MARKING GUIDELINE



2. ASSEMBLY
1 MARK FOR EVERY PART CORRECTLY ASSEMBLED
(10 PARTS - 1) = 9



FOR OFFICIAL USE ONLY	
INCORRECT ORTHOGRAPHIC PROJECTION	
INCORRECT OVERALL SCALE	
INCORRECT HATCHING	
PARTS NOT ASSEMBLED	
TOTAL PENALTIES (-)	

ASSESSMENT CRITERIA					
RIGHT VIEW					
		POSSIBLE	OBTAINED	SIGN	MODERATED
1	HOUSING + SYMMETRY	6			
2	COVER + SCREWS	4			
3	FLANGE COUPLING	2			
4	M16 NUT + SHAFT	4			
SUBTOTAL		16			
SECTIONAL FRONT VIEW					
1	HOUSING	10 1/2			
2	COVER	4 1/2			
3	M10 SCREW	7			
4	SPUR GEAR	9			
5	BALL BEARING	2 1/2			
6	SHAFT + KEY + BUSH	15 1/2			
7	FLANGE COUPLING	9			
8	M16 NUT	4			
SUBTOTAL		62			
GENERAL					
1	CENTRE LINES	3			
2	ASSEMBLY	9			
SUBTOTAL		12			
TOTAL		90			
PENALTIES (-)					
GRAND TOTAL					